*Clean Energy*, 2019, Vol. 3, No. 3, 227–234

doi: 10.1093/ce/zkz025
Homepage: https://academic.oup.com/ce

Research Article

# A case study of digital-twin-modelling analysis on power-plant-performance optimizations

Bin Xu[1,*] June Wang[2], Xinping Wang[2], Zhihong Liang[3], Liming Cui[3], Xiao Liu[1] and Anthony Y. Ku[1]

[1]China Energy National Institute of Clean-and-Low-Carbon Energy, Future Science City, Changping District, Beijing, China;
[2]China Energy Qinhuangdao Power Generation Co. Ltd Qinhuangdao City, Hebei Province, China;
[3]China Energy Fossil Fuel Power Generation Operation Management Center, Fuchengmen, Beijing, China;

*Corresponding author. E-mail: bin.xu.ck@nicenergy.com

## Abstract

China Energy's National Institute of Clean-and-Low-Carbon Energy (NICE) is developing a Power Plant Smart Management (PPSM) platform that employs digital-twin technology to undertake techno-economic modelling analysis on China Energy's existing coal-fired power-plant units and explore cost-effective solutions to improve those plant units' thermal efficiencies and operating performance. This paper presents a case study of PPSM on a 320-MWe coal-fired thermal power-plant unit, demonstrating how the digital-twin technology was employed to explore and analyse optimization solutions. Various optimization solutions and their cost-effectiveness were assessed using the digital-twin-modelling analysis; the results indicated the optimization solutions are expected to improve the plant unit's operating efficiency and reduce its current electricity-generation coal consumption by up to 3.5 g/kWh standard coal equivalent (sce), worth annual fuel-cost savings of approximately 4 million RMB for a single unit or 8 million RMB for the two identical 320-MWe units that the power plant currently operates. The digital twin was also employed to assess the power-plant unit's operating economics during both summer and winter. In summer, when the unit operates in electricity-generation-only mode, the unit's operating thermal efficiency could drop by up to 6% points following the grid demand of load changes from 100% maximum continuous rating (MCR) down to 30%MCR, resulting in an ~ 45 RMB/MWh increase of electricity-generation cost. In winter, when the unit operates in combined heat and power (CHP) cogeneration mode, for the same boiler load, the CHP operation increases the plant unit's operating profit with increasing district-heating duty, although the relative profit gain from the CHP cogeneration could start to decrease when the district-heating steam-extraction flow increases to a certain point that varies depending on the market prices of heat and electricity, while the fuel cost was found to be equivalent to ~50% of the unit's total CHP income cogenerated from its electricity and district heat outputs.

**Keywords:** thermal power plant; smart management; digital twin; modelling; simulation; performance optimization

**Received:** 4 May 2019; **Accepted:** July 26, 2019

© The Author 2019. Published by Oxford University Press on behalf of National Institute of Clean-and-Low-Carbon Energy. This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (http://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact journals.permissions@oup.com



Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

## Introduction

The electric-power-generation market in China is highly regulated by the central government and the regional administrations. In order to meet the power grid's demand to accommodate the increasing intermittent power supplies from renewables such as wind and solar power, many baseload coal-fired power-generation units are required to operate in a load-following mode. This not only causes undesirable thermal stresses to the plants' pressure parts and turbomachinery, but also results in low thermal efficiencies and low-capacity factors that impair the economics of those power plants. It was reported that >50% of coal-fired power plants in China operated at losses in year 2018 due to the high fuel cost and depressed electricity or heat prices [1, 2]. The best way for those coal-fired power plants to stay competitive or become more profitable is to improve their operating efficiencies to reduce the plant-operating costs, of which 60~70% generally comes from fuel.

China Energy Investment Corporation Limited ('China Energy') is the world's largest energy and power company, with 239-$GW_e$ installed power-generation capacity including 176-$GW_e$ capacity from its fleet of 478 coal-fired power-generating units [3]. The coal-fired power-plant fleet generated 760 billion kWh electricity and approximately 600 million tons of $CO_2$ emission in year 2017. The average electric-generation coal consumption is approximately 305 g(sce)/kWh [4]—a 1-g(sce)/kWh coal-consumption reduction for the fleet would be equivalent to an annual fuel-cost reduction of 450 million RMB for China Energy. A 1% point improvement in the average thermal efficiency would reduce the fleet's annual fuel cost by approximately 1.4 billion RMB and $CO_2$ emissions by 6 million tonnes.

Aiming to optimize the operations of China Energy's existing coal-fired power-plant fleet, China Energy's National Institute of Clean-and-Low-Carbon Energy (NICE) is developing a Power Plant Smart Management (PPSM) platform. The core of the PPSM platform is its customized 'digital-twin' numerical models of the existing coal-fired power-generation units. Digital twins can be employed to simulate and analyse each unit's actual operating performance in comparison to its optimal performance at varying operating conditions, investigate the root causes of the performance deviations and evaluate the cost-effectiveness of optimization solutions. This paper presents a case study for a 320-MWe coal-fired thermal power-plant unit. A digital twin was built for the power-plant unit including its boiler island, steam-turbine island and the emission-control sub-systems. The digital twin was employed to simulate the unit's actual operating performance in comparison with its original equipment manufacturer (OEM) design performance, analyse the impacts of varying operating conditions on plant thermal performance and evaluate the impacts of fuel, electricity and heat market-price variations on plant economics.

## 1 Methods

### 1.1 Digital-twin model

The case study was to build a digital-twin numerical model of a coal-fired thermal power-plant unit and use the digital twin to analyse the plant-operating performance and explore optimization solutions to enhance the plant-operating economics. A digital twin is a comprehensive physics-based numerical model of a power-plant unit that can simulate the thermodynamic performance and calculate the heat and mass balance of the plant's overall system, sub-systems and major plant components, respectively, for any given operating conditions. The digital twin developed using the Thermoflow™ software suite included models of the boiler island, steam-turbine island and emission control equipment as shown in Fig. 1. Emission control equipment consists of selective catalytic reduction (SCR), electrostatic precipitator (ESP), and sea-water flue gas desulfurization (SWFGD) scrubber. The digital twin was configured and sized using the unit's OEM design specification data, which represent the unit's OEM design performance at 320-MWe base load.

The digital twin was then calibrated using the plant unit's recent operating-performance data extracted from the distributed control system system, which represents the actual performance of the thermal power-plant unit. The calibrated digital model was validated by simulating the performances of various part-load-operating cases that were well in line with the actual performances. Therefore, the calibrated digital twin can be employed to simulate plant performance under given operating conditions.

### 1.2 Power-plant design and operating data

The power-plant unit selected for the case study was a subcritical coal-fired thermal power-plant unit with main design performance data summarized in Table 1. The thermal power-generation unit has an electricity-generation capacity of 320 MWe with conditions for the high pressure (HP) and intermediate pressure (IP) steam as follows: 16.7 MPa (HP)/538°C (HP)/538°C (IP); it was designed for combined heat and power (CHP) cogeneration with a steam-extraction capacity of 350 tonnes/hr from its steam-turbine IP and low pressure (LP) crossover for district heating during wintertime.

Data on plant operations and economics were provided by the plant operator. The operating data reported in this paper correspond to 'representative' conditions during a period of steady-state operation near full load in the summer and winter of 2018. Variability in the data due to load following was observed, but a discussion of its sources and the performance of the digital twin under load-following conditions is beyond the scope of this paper.

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

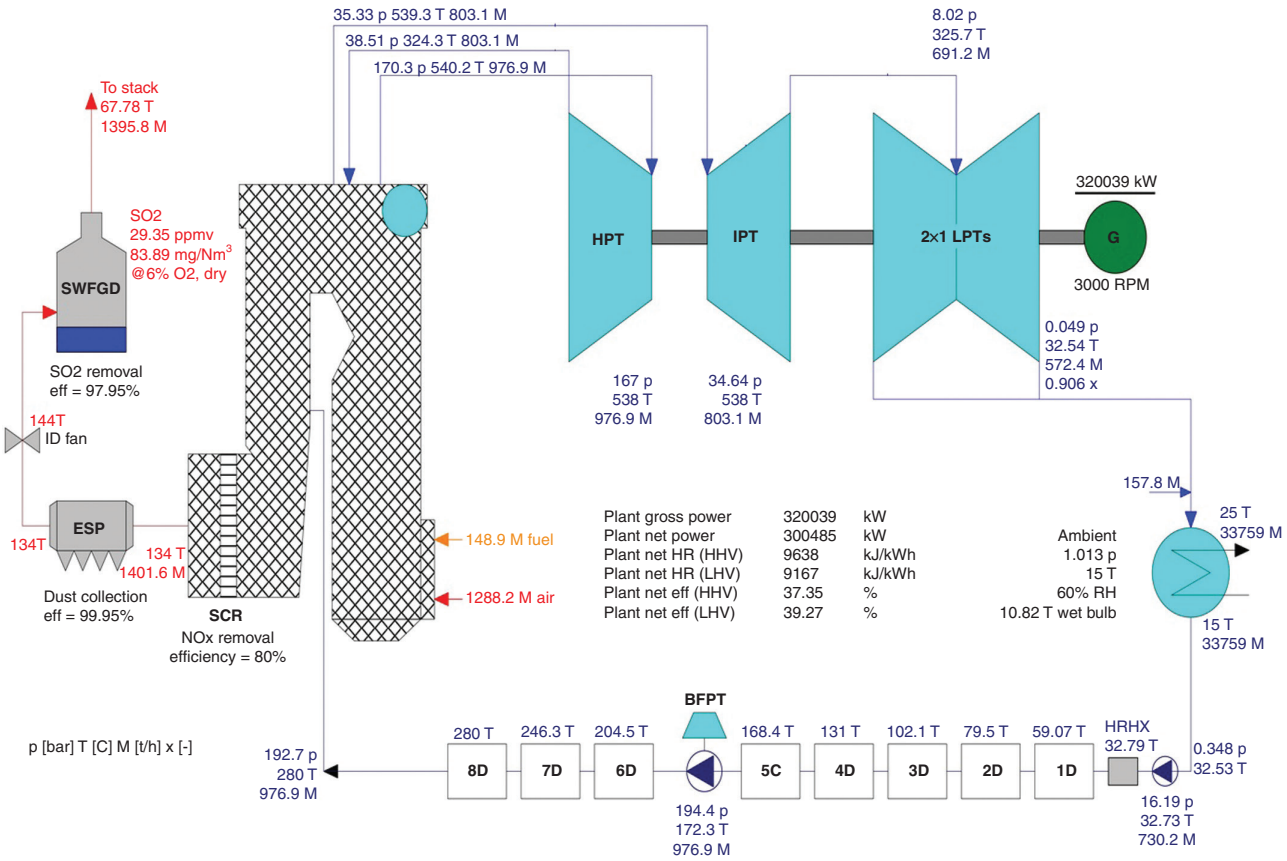



**Fig. 1** The digital twin of a 320-MWe coal-fired thermal power plant—baseline design performance

**Table 1** Design performance data of a 320-MWe coal-fired thermal power-plant unit

| Design load | Gross electric output | HP inlet steam | | | IP inlet steam | | Feed-water temperature | Condenser pressure | District-heating steam-extraction flow |
|---|---|---|---|---|---|---|---|---|---|
| | MWe | Tonnes/hr | MPa | °C | MPa | °C | °C | kPa | Tonnes/hr |
| THA | 320.0 | 976.9 | 16.7 | 538 | 3.464 | 538 | 280 | 4.9 | – |
| TMCR | 329.6 | 1013.6 | 16.7 | 538 | 3.583 | 538 | 282.4 | 4.9 | – |
| CHP | 253.6 | 1013.6 | 16.7 | 538 | 3.571 | 538 | 282.3 | 4.9 | 350 |

THA, turbine heat-rate acceptance; TMCR, turbine maximum continuous rate; CHP, combined heat and power.

## 2 Results and discussion

### 2.1 Power-plant-performance analysis and optimization solutions

Fig. 2 presents the schematic diagram of the digital twin of the power-plant unit operating at full load (320 MWe) in the summertime of year 2018.

Table 2 compares the actual operating conditions and performance of the plant at full load (Fig. 2) to that of the OEM design full load (Fig. 1). In actual operation, the power plant unit achieved its targeted HP and IP steam temperatures of 538°C. However, there was ~3.5% more HP steam flow, 2.5-bar lower HP steam pressure, 5°C lower boiler feed-water temperature and ~1.46% points lower thermal efficiency compared with the design. Some of the deviations were due to non-operational factors such as the higher-than-design ambient and cooling-water temperatures resulting in higher condenser pressure, while some were due to operational factors such as poor operating practices or hardware components' performance deterioration.

One factor significantly impairing the unit's operating thermal efficiency was the excessively high boiler exit (i.e. air-heater exit) flue-gas temperatures. The actual operating temperatures reached an average of 147°C and as high as 160°C at some points; this was much higher than the 134°C of the design case.

The root cause of this effect was identified by the digital-twin-modelling analysis. It was revealed that a large fraction (up to nearly 50%) of the cold primary air stream into the air heaters was bypassed and used as tempering air to control the mills' outlet temperatures; this resulted

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

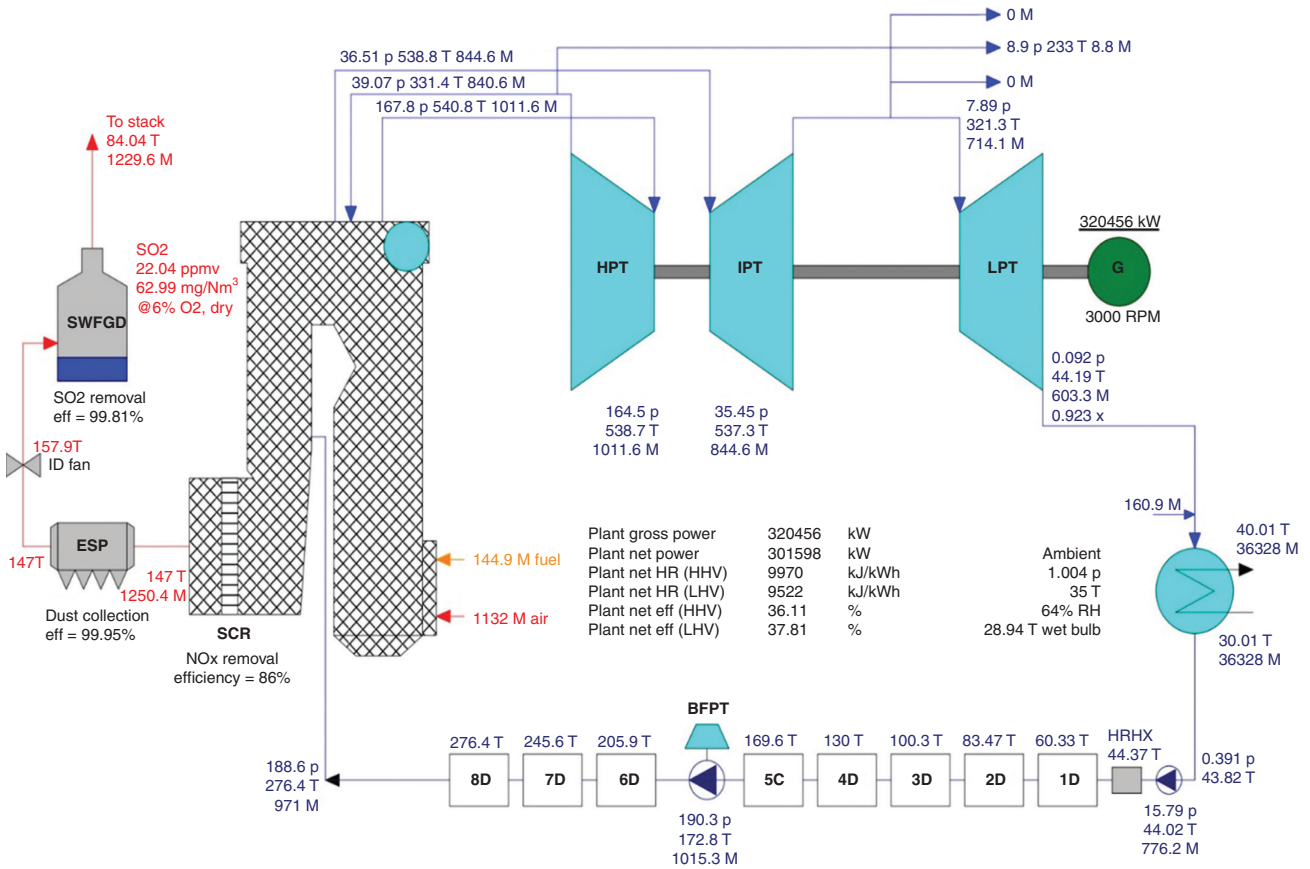



**Fig. 2** The 320-MW power-plant-unit digital twin—actual full-load operating performance

**Table 2** Comparisons of plant performances (actual vs. design)

| | | Design | Operation | Deviation | Notes |
|---|---|---|---|---|---|
| Plant gross output | $MW_e$ | 320.04 | 320.46 | 0.13% | |
| Plant electric efficiency (gross) | %LHV | 41.83% | 40.17% | –1.66%pts | |
| Plant electric efficiency (net) | %LHV | 39.27% | 37.81% | –1.46%pts | |
| HP steam flow | tph | 976.9 | 1011.6 | +3.55% | Operational |
| HP steam temperature | °C | 538 | 538.7 | +0.1% | Operational |
| IP steam temperature | °C | 538 | 537.3 | +0.1% | Operational |
| Boiler feed-water temperature | °C | 280 | 276.5 | –1.25% | Operational |
| HP steam pressure | MPa | 16.70 | 16.45 | –0.25 MPa | Operational |
| IP steam pressure | MPa | 3.46 | 3.55 | +2.6% | Operational |
| Condenser pressure | MPa | 0.049 | 0.092 | +0.05 MPa | Operational |
| Air-heater exit-gas temperature | °C | 134 | 147 | 13°C | Hardware |
| Ambient temperature | °C | 15 | 38 | +23°C | Uncontrollable |
| Cooling-water temperature | °C | 15 | 30 | +15°C | Uncontrollable |

in reduced heat transfer to the primary air stream from the flue-gas stream. This situation was more a hardware-design issue rather than an operating problem. Bypassing cold primary air for mill-tempering control is common design practice for conventional pulverized coal-fired power plants, and it is common for plants to have oversized air heaters in order to provide enough margin to maintain its mills' air-tempering capability and flexibility to cope with firing different types of coals and variations in loads and operating conditions.

The minimum allowable air-heater exit-flue-gas temperature is normally dictated by the acid dew point of the flue gas, depending on the sulfur and moisture contents of the coal fired in the boiler. For the type of coal currently used by the thermal power unit under study, the air-heater exit-flue-gas temperature can be as low as 115°C, without incurring acid corrosion risk in the air heater; reducing the boiler exit-gas temperature to as low as is allowed would significantly improve the boiler thermal efficiency and reduce the electricity-generation fuel-consumption rate.

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

The digital twin was employed to investigate various optimization solutions within the boiler island to reduce the air-heater exit-flue-gas temperature without modification of the existing air heaters. An optimization solution was identified (i.e. Opti 1 in Table 3) to be able to reduce the boiler exit-flue-gas temperature by up to 25°C from its current level, leading to an improvement in the plant unit's heat rate by ~86 kJ/kWh, or equivalent to an ~2.6-g(sce)/kWh reduction of electricity coal consumption. The modelling analysis also showed that this optimization solution can be flexibly operated either to maximize electric output during peak time or maximize fuel-cost saving during off-peak time for the best economic gain based on the plant operator's preference.

A second optimization solution (Opti 2 in Table 3) was identified within the steam-turbine island, which optimizes the utilization of the heat of the turbine bleed steam going to the dearator. This solution has the potential to increase the plant heat rate by up to 15 kJ/kWh, or equivalent to a 0.5-g(sce)/kWh reduction in coal consumption.

The combination of the two optimization solutions (Opti 1+Opti 2) can potentially reduce the baseline electricity coal consumption from its current level by up to 3.5 g(sce)/kWh, or equivalent to an income increase of 875-1075 RMB/hour for the same fuel consumption of its current 320-MWe full load, based on the coal (sce) price of 600 RMB/tonne and electricity price of 0.372 RMB/kWh.

## 2.2 Effects of load following

Due to power-generation overcapacity across the country, coal-fired power-plant operators in China are encouraged to pursue load-following flexibility with the lowest possible load-operating capability of their existing power-generation units. Not all operators are aware that their units were originally designed and optimized for baseload operations and that frequent irregular load changes cause undesirable thermal stresses to the plants' pressure parts and turbomachinery, resulting in higher maintenance costs. A long period of low-load operation also incurs significant penalties in thermal efficiency and low-capacity factors increase operating costs. For example, a 1000-MWe unit operating at 50%load or 40%load would increase the electricity-generation coal consumption by 41g(sce)/kWh or 68g(sce)/kWh, respectively, compared with that at full load [5].

Digital-twin simulations can assist power-plant operators to gain better understanding in the techno-economic impacts of various operating conditions on their plant units and provide guidance for optimizing operating practices over the long term. Specifically, the digital-twin models can be used to assess the impacts of variations in operating conditions on the plant thermal performance and the impacts of the market (price) fluctuations of fuel, electricity and district heat on the plant-operating economics.

Many baseload coal-fired power-generation units are required by the grid to do frequent load following in order to accommodate intermittent power generation from renewable sources such as wind and solar. Fig. 3 shows the results from simulations of the 320-MWe power-plant performance under a turndown scenario from 100% maximum continuous rating (MCR) down to 30%MCR. The electric efficiency drops by ~6% points—equivalent to an increase in the coal-consumption rate of 50 g (sce)/kWh, resulting in a 45-RMB/MWh increase in the electricity-generation cost. Use of the digital-twin models to reduce these impacts will be the subject of future work.

## 2.3 CHP-operating strategies and seasonal optimization opportunities

The 320-MWe thermal power-plant unit under study also provides district heating during wintertime through steam extraction from its IP and LP crossover by up to 350 tonnes/hr and up to 40 tonnes/hr steam export from its HP turbine exhaust for local industry process utilization. CHP

**Table 3** Optimization-solution impacts on performance and economics

| | | | 320 MWe (electricity-generation mode) | | | | |
|---|---|---|---|---|---|---|---|
| | | | Opti 1 | | | Opti 1 + Opti 2 | |
| Optimization-solution assessment | | Baseline[a] | Operation mode 1[b] | Operation mode 2[c] | Opti 2 | Operation mode 1[b] | Operation mode 2[c] |
| Gross electricity coal-consumption reduction | g/kWh | 306.3 | –2.22 | –2.59 | –0.48 | –3.10 | –3.45 |
| Extra power output | MWe | 0.00 | 2.38 | 1.02 | 0.40 | 2.51 | 1.15 |
| Fuel-cost difference (@600RMB/T) | RMB/hr | 0.00 | 8 | –308 | –20 | –141 | –447 |
| Additional electricity income (@0.372RMB /kWh) | RMB/hr | 0.00 | 885 | 379 | 147 | 934 | 427 |
| Total hourly-income increase | RMB/hr | 0.00 | 877 | 687 | 167 | 1075 | 874 |

[a]320-MWe full-load comparison based on maintaining the same main steam conditions at the steam-turbine HP inlet.
[b]Peak-time-operation mode targeting for high profit.
[c]Off-peak-time-operation mode targeting for low-fuel-consumption rate.

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

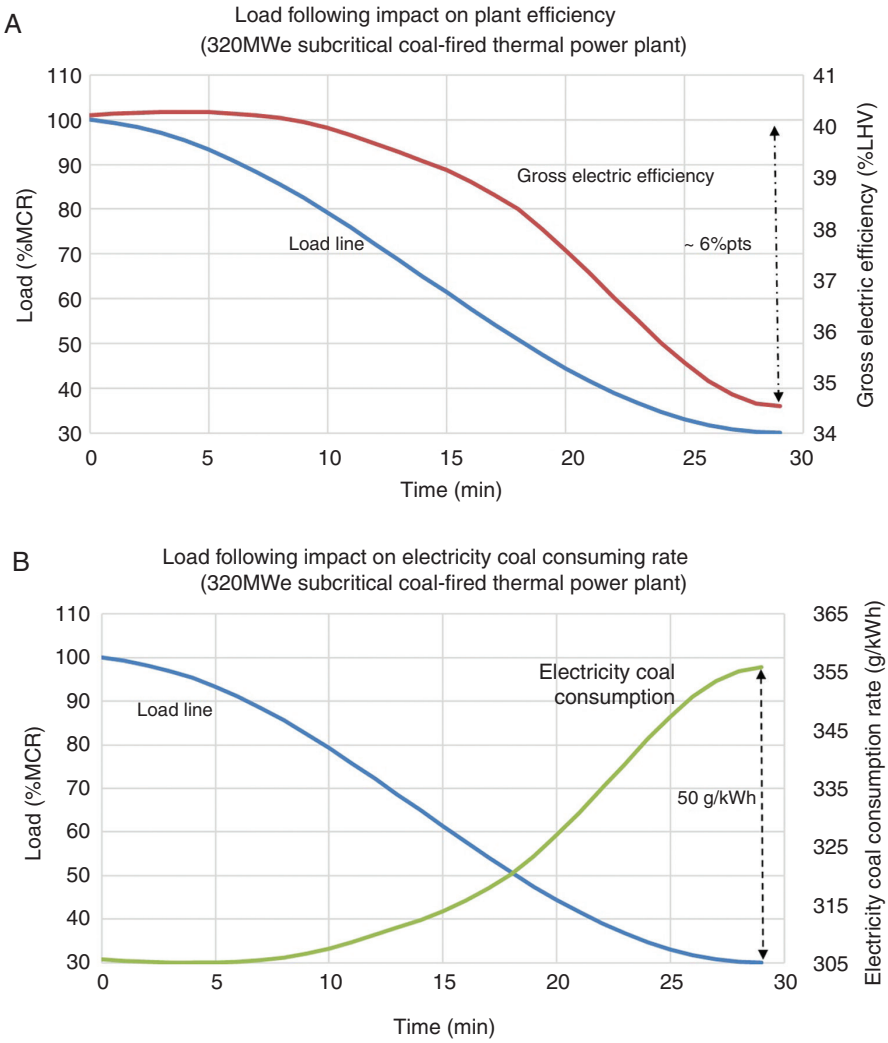



**Fig. 3** Effect of load following on (a) power-plant electric efficiency and (b) electricity coal consumption

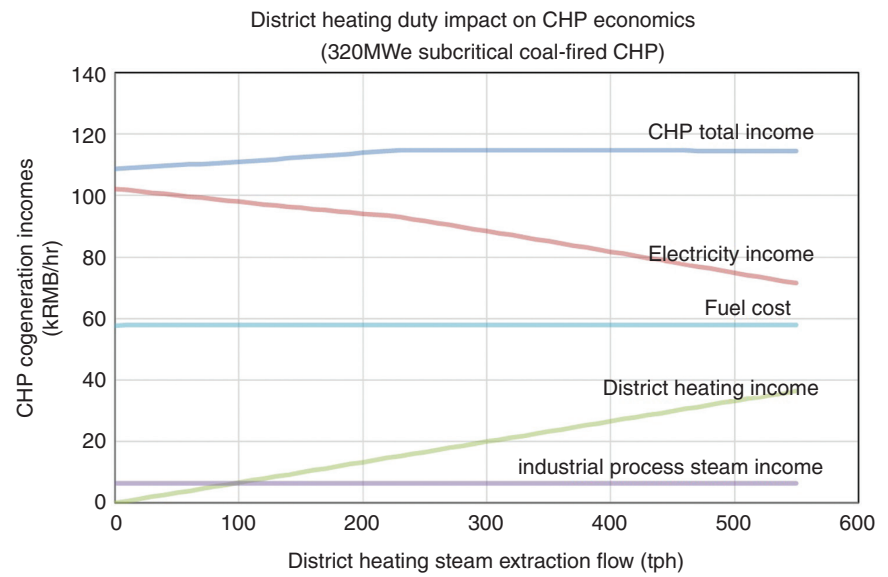



**Fig. 4** District-heating steam-extraction impact on plant CHP cogeneration economics

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

cogeneration generally improves the plant unit's overall CHP efficiency and operating economics, with benefits increasing with district-heating duty.

Fig. 4 presents the impact of the district-heating steam-extraction flow rate on the plant CHP cogeneration economics simulated by the digital twin. The total income of the power-plant unit's CHP operation came from three components, namely the electricity production (0.372 RMB/kWh), district heating (25 RMB/GJ) and industrial-process steam (65 RMB/GJ). The simulations assumed the power plant running in CHP cogeneration mode with the same amount of fuel fired as that of 320-MWe full-load operation with a fixed small amount of steam export for the industrial process. Increasing the district-heating steam-extraction flow increases the income from district heating, while simultaneously decreasing the income from electricity production. The fuel cost was found to be ~50% of the total CHP income. The net impact is increased revenues due to the favourable relative pricing for heat. However, this trend breaks down at high levels of steam extraction, where the performance of the LP steam turbine suffers due to inadequate steam mass flow.

Fig. 5a and 5b show the results of sensitivity analysis based on the market prices of heat and electricity, respectively. The inflection in the curves corresponds to the steam-extraction rate where debits to the efficiency of the steam turbine become increasingly severe.

At the current heat pricing of 25 RMB/GJ, the relative benefit of CHP cogeneration will increase until the district heat output reaches ~175 $MW_{th}$. Then, the benefit turns relatively flat until the district-heating duty reaches ~300 $MW_{th}$, when the CHP benefit starts to decrease following a further increase in the district-heating duty. The reversing trend could be avoided when the heat price increases to >26 RMB/GJ or the electricity price drops to <0.36 RMB/kWh.

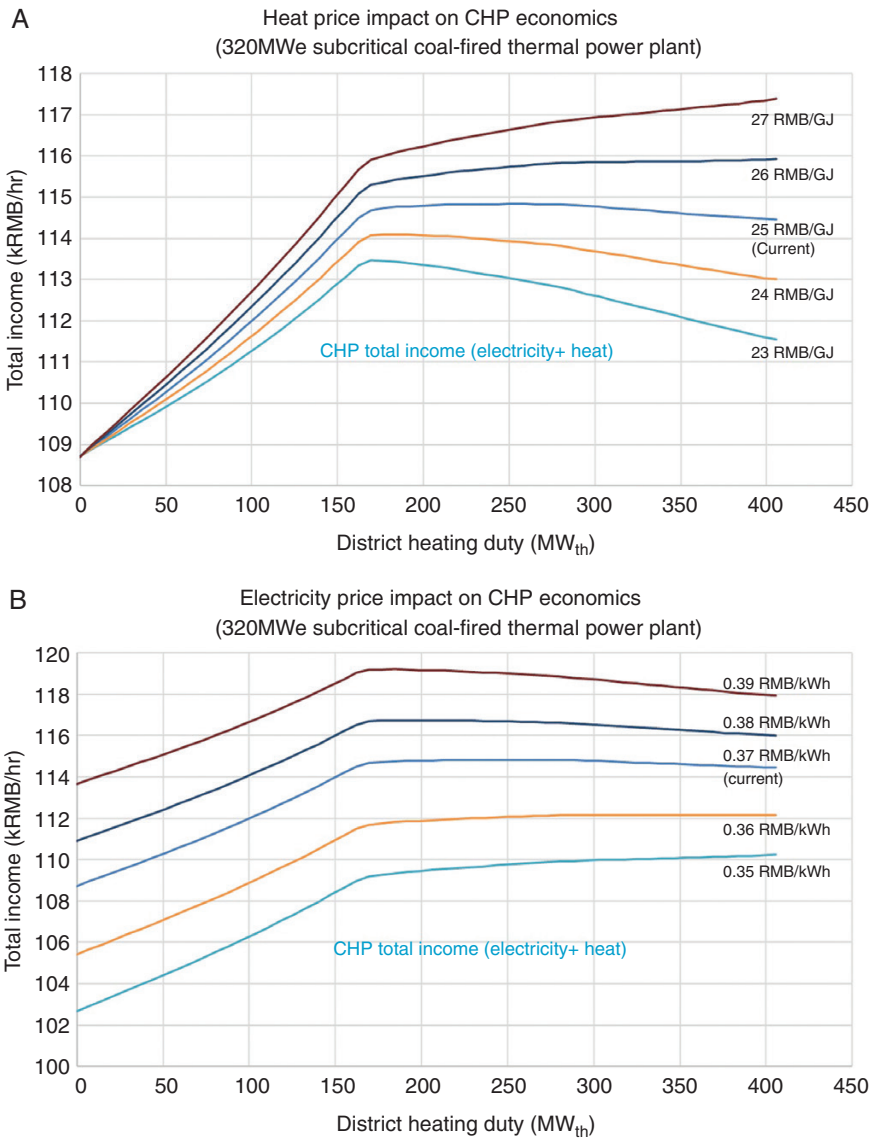



**Fig. 5** : (a) Heat-price impact on CHP economics and (b) impact of electricity prices on CHP-operating economics

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019

These sensitivity curves can be used by plant operators to maximize the revenue of the plant, subject to additional operating directives issued by relevant market authorities.

## 3 Conclusions

The case study presented in this paper demonstrates how a digital twin can be built for a power-plant unit and employed as an effective tool to simulate the thermodynamic performance of the power-plant unit operating under various operating conditions, analyse the root causes of performance deviations, and explore optimization solutions and assess their cost-effective impacts on plant-operating economics.

It should be noted by both plant operators and power-market regulators that frequently irregular load changes to a coal-fired power plant cause undesirable thermal stresses to the plants' pressure parts and turbomachinery, resulting in a shortened lifespan and increasing operation and maintenance (O&M) costs in the long term; low-load operation also incurs significant penalty in thermal-efficiency loss and low-capacity factors impairing long-term operating economics. For a 320-MWe unit, reducing the load from 100%MCR down to 30%MCR can result in ~6% points efficiency loss, equivalent to a 50-g(sce)/kWh coal-consumption increase or 45-RMB/MWh increase in the electricity-generation cost.

CHP cogeneration at a thermal power plant generally improves the plant unit's overall CHP efficiency and operating economics following the increase in district-heating duty, which, however, could reach a turning point when the relative CHP economic gain may start to decrease; the turning point varies largely depending on the heat price and electricity price on the market. A digital twin of a power-plant unit can be an effective tool to assist the plant operators to operate their plant units in a cost-effective way following the price fluctuations of heat and electricity in the market.

## Acknowledgements

The authors would like to thank Mr Zhang Xiang-an, Mr Zhan Guo-bin, Mr Li Yin-gui and Ms Guo Yan of China Energy Qinhuangdao Power Generation Co. Ltd for their support in plant-data acquisition and constructive discussions during this study.

## Conflict of Interest

None declared.

## References

[1] Fossil fuel fired power generation sector is in full loss. http://www.xinhuanet.com/power/2018-02/02/c_1122354693.htm (13 October 2019, date last accessed).

[2] Coal-fired power industry is facing many uncertainties. http://www.xinhuanet.com/fortune/2018-07/18/c_1123140710.htm (13 October 2019, date last accessed).

[3] China Energy Investment Corporation (China Energy) company profile. http://www.ceic.com/gjnyjtwwEn/jtgk/chnjtjs.shtml (13 October 2019, date last accessed).

[4] Interpretation of the major economic and technical indicators of the New Five Power Generation Group in 2018. http://www.cpnn.com.cn/zdzg/201903/t20190304_1123639.html (13 October 2019, date last accessed).

[5] Shu-hua Mi. Clean low carbon development of China Energy group's coal-fired power generation fleet. In: China Electricity Council's 2019 Economic Situation and Power Development Seminar, 18 April 2019. https://www.nengapp.com/news/detail/2791708 (13 October 2019, date last accessed).

Downloaded from https://academic.oup.com/ce/article-abstract/3/3/227/5614342 by guest on 08 November 2019